

Lateral Movement Inference in Evolving Enterprise Networks via High-order Temporal Graph Learning

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Abstract—Lateral Movement (LM) is a critical stage of advanced persistent threats (APTs), during which adversaries propagate within enterprise networks to expand their access and compromise high-value assets. Such behaviors are inherently stealthy, as attackers leverage legitimate credentials and mimic benign activities, making detection particularly challenging. Recent advances in temporal graph learning address this problem by modeling evolving interactions and enabling inductive inference over partially observed systems. However, existing approaches predominantly rely on pairwise interaction modeling, implicitly assuming that complex behaviors can be decomposed into independent relations. This assumption is often violated in real-world enterprise environments, where interactions are inherently high-order and involve multiple entities within a single event, leading to structural ambiguity in temporal representations. To address this limitation, we propose a high-order temporal modeling framework that captures event-level interactions among multiple entities, enabling more expressive representations of complex dependencies in evolving enterprise networks. Furthermore, we revisit the attention mechanism in temporal graph models and introduce a smooth normalization strategy that improves the ability to distinguish relevant interactions from noise while preserving multiple meaningful dependencies. Extensive experiments under inductive temporal link prediction settings demonstrate that our approach consistently outperforms strong temporal graph-based baselines and provides more robust modeling of complex temporal interactions.

Index Terms—Lateral Movement, Temporal Graph Learning, High-order Interaction Modeling, Evolving Enterprise Networks

I. INTRODUCTION

Lateral Movement (LM) represents a critical phase of advanced persistent threats (APTs), during which adversaries progressively traverse enterprise networks to expand their foothold and gain access to high-value assets [1], [2]. Unlike external attacks, LM often unfolds through legitimate system interactions, where attackers exploit compromised credentials and mimic benign user behavior [3]. As a result, such activities typically leave minimal distinguishable traces in system and

network logs, making them particularly difficult to detect. In practice, enterprise environments are highly dynamic and heterogeneous, involving continuously evolving interactions among users, hosts, applications, and services. This complexity further obscures malicious behaviors, as legitimate and adversarial activities are often intertwined within the same operational context. Consequently, traditional detection systems, including signature-based and rule-based approaches such as security information and event management (SIEM) and network intrusion detection systems (NIDS), are often limited to identifying simplistic or previously observed attack patterns, while failing to capture sophisticated and evolving attack strategies [4]. These challenges highlight the need for more expressive modeling approaches capable of capturing complex and evolving interaction patterns in enterprise networks. As illustrated in Fig. 1, lateral movement attacks typically involve multi-step propagation across multiple entities, forming complex temporal dependencies that are difficult to distinguish from benign behaviors.

Recent advances have increasingly turned to graph-based and temporal learning methods for modeling interactions in enterprise networks, where authentication events and system activities are represented as dynamic relational data [5]–[7]. In particular, temporal graph learning has demonstrated strong potential in capturing evolving dependencies and enabling inference over partially observed systems. Prior work has further explored these ideas in the context of lateral movement detection, including temporal link prediction frameworks that model anomalous interactions in evolving networks [8], as well as inductive temporal graph models designed to handle dynamic and partially observable enterprise environments [9].

Despite the significant progress achieved at both the data representation and modeling levels, existing approaches predominantly rely on pairwise interaction modeling, implicitly assuming that complex behaviors can be decomposed into independent relations. Such an assumption is often vi-

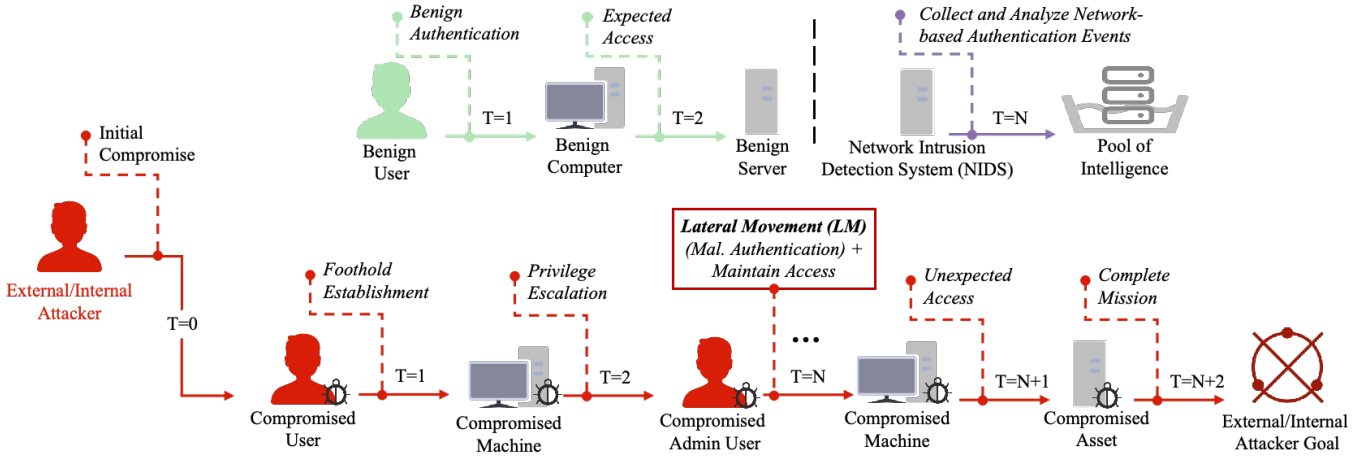


Fig. 1. Illustration of authentication event evolution in enterprise networks. The upper trajectory shows benign behaviors, while the lower trajectory depicts a lateral movement (LM) attack involving multi-step compromises. This highlights the importance of modeling multi-hop and high-order temporal dependencies.

olated in practice, where interactions are inherently high-order and involve multiple entities within a single event, such as coordinated authentication processes or multi-stage access patterns. Decomposing these interactions into pairwise relations inevitably leads to the loss of structural information, resulting in ambiguity in temporal representations. In addition to these structural limitations, attention mechanisms employed in temporal graph models typically rely on SoftMax-based normalization to aggregate historical dependencies. While widely adopted, SoftMax is inherently constrained by a trade-off between concentration and diversity, limiting its expressiveness in capturing complex dependency patterns [10]. Specifically, overly concentrated attention distributions tend to suppress multiple relevant interactions, whereas overly smooth distributions assign non-negligible weights to less informative events. As a result, SoftMax struggles to simultaneously suppress noise and preserve multiple meaningful signals, thereby reducing the discriminative power of temporal attention. Taken together, these observations indicate that the limitations of existing temporal graph models stem not only from data representation, but also from the expressiveness of the attention mechanism itself. In particular, temporal attention suffers from both structural ambiguity induced by pairwise modeling and numerical ambiguity introduced by SoftMax normalization.

In this work, we address the above limitations through a unified framework that refines temporal attention from both structural and numerical perspectives. To resolve the structural ambiguity introduced by pairwise interaction modeling, we adopt a high-order temporal modeling framework that captures event-level interactions among multiple entities, enabling multiple entities involved in a single event to be modeled jointly rather than decomposed into independent relations. This allows for a more faithful representation of complex and coordinated behaviors in evolving enterprise networks. To further mitigate the numerical ambiguity induced by SoftMax normalization, we revisit the design of the attention weighting mechanism. Instead of relying on a single-mode normalization

that is constrained by the concentration–diversity trade-off, we incorporate a smooth normalization strategy that enables the preservation of multiple salient dependencies while effectively suppressing low-relevance noise. This enhances the discriminative capability of temporal attention in capturing diverse and concurrent interaction patterns.

By integrating high-order interaction modeling with an improved attention normalization mechanism, our approach provides a two-level refinement of temporal attention, enhancing both structural expressiveness and numerical discriminability. As a result, the proposed framework enables more accurate and robust inference of lateral movement in dynamic and partially observed enterprise networks. Based on this design, our main contributions are as follows:

- We propose a high-order temporal modeling framework that captures multi-entity interactions at the event level, enabling a more expressive representation of complex multi-entity behaviors in evolving enterprise networks.
- We develop a smooth attention normalization mechanism that alleviates the limitations of SoftMax normalization, enhancing the ability to preserve multiple relevant dependencies while suppressing low-relevance noise.
- We conduct extensive experiments on the same real-world enterprise network dataset used in prior work and demonstrate that our approach achieves promising results for lateral movement inference, including under inductive and partially observed settings.

II. RELATED WORK

A. Lateral Movement Detection in Enterprise Networks

Lateral Movement (LM) detection has been extensively studied in the context of enterprise network security, where the goal is to identify adversarial propagation across hosts, users, and services. Early approaches primarily relied on host-based monitoring and provenance tracking systems [11], [12], which provide high-fidelity insights into system-level activities but often suffer from scalability limitations and restricted global

visibility. Network-based methods have also been explored to analyze communication patterns and detect anomalous behaviors, yet they typically lack fine-grained contextual understanding of authentication events. More recently, graph-based approaches have emerged as an effective paradigm for modeling LM behaviors, where entities such as users, hosts, and applications are represented as nodes, and their interactions are captured as edges. Prior work includes both temporal graph-based methods and system-level modeling approaches [8], [9], [13]. In particular, temporal graph-based methods have shown strong potential in capturing evolving attack patterns by modeling authentication events as dynamic interactions. Temporal link prediction frameworks [8] have been proposed to identify anomalous connections in evolving networks, while inductive temporal graph models [9] further enable generalization to previously unseen nodes and partially observed environments. While these approaches demonstrate the effectiveness of graph-based modeling for LM detection, they primarily rely on pairwise interactions and do not fully capture complex multi-entity behaviors that arise in real-world enterprise networks.

B. Temporal Graph Learning for Dynamic Systems

Graph Neural Networks (GNNs) have become a fundamental tool for modeling relational data, enabling the learning of node representations through message passing over graph structures [5]. Building upon this paradigm, recent work has extended GNNs to dynamic settings, where interactions evolve over time. Temporal graph learning models, such as Temporal Graph Networks (TGN) [6] and Temporal Graph Attention (TGAT) [7], incorporate temporal information into message passing and attention mechanisms, allowing for the modeling of continuous-time interactions and evolving dependencies. These advances have been further applied to dynamic link prediction and anomaly detection tasks, where the goal is to infer future or suspicious interactions based on historical observations. For instance, temporal link prediction frameworks have been proposed to detect anomalous connections in evolving networks [8], while inductive temporal graph models enable generalization to unseen nodes and partially observed environments [9]. Such capabilities are particularly important in enterprise security settings, where network structures continuously evolve and complete observability is rarely available. Despite their success, most existing temporal graph learning approaches fundamentally rely on pairwise interaction modeling, where complex events are decomposed into independent edges between pairs of entities. However, real-world systems often exhibit higher-order interaction patterns that cannot be fully captured by pairwise relations [14]. Higher-order interactions have been extensively studied in hypergraph neural networks [15], [16], and recent work has further shown that incorporating such structural information can significantly improve link prediction performance in temporal graphs [17]. In enterprise networks, authentication events often involve multiple entities simultaneously, forming complex interaction patterns that cannot be faithfully represented through pairwise

decomposition. As a result, such modeling choices may lead to incomplete structural representations and hinder the accurate characterization of evolving behaviors. In addition to structural modeling, attention mechanisms play a crucial role in temporal graph learning. Existing methods typically adopt SoftMax-based attention [18], while alternative normalization strategies such as Sparsemax have been proposed to improve sparsity and interpretability [19]. More recently, MultiMax normalization has been proposed to better preserve multiple salient modes in attention distributions [10]. Nevertheless, these approaches still face challenges in balancing the preservation of multiple relevant dependencies and the suppression of noise, which motivates further exploration of more expressive attention normalization schemes.

III. METHOD

A. Problem Definition

We consider lateral movement (LM) inference in evolving enterprise networks, where authentication interactions occur dynamically over time among heterogeneous entities. Let $\mathcal{V}$ denote the set of entities in the network, including users, hosts, applications, and services, and let $\mathcal{E} = \{e_1, e_2, \dots, e_T\}$ denote the sequence of timestamped authentication events. Each event e_t occurs at time t and involves a subset of participating entities, rather than a single pair of nodes. This distinction is important in enterprise environments, where a single authentication event may simultaneously encode multiple roles and dependencies.

Following prior temporal graph-based approaches, we formulate LM inference as a temporal link prediction problem. Specifically, given the partially observed interaction history up to time t , denoted as $\mathcal{E}_{\leq t}$, the goal is to estimate the likelihood of future or unseen interactions between entities, including those involving previously unseen nodes. Formally, the model learns a scoring function $f(u, v, t)$ that predicts the probability of an interaction between entities u and v at time t . Unlike conventional pairwise graph formulations, however, we do not assume that each event can be faithfully decomposed into independent binary relations. Instead, we explicitly account for the high-order nature of authentication events by leveraging the full set of participating entities as contextual information. Reducing such events to pairwise edges may discard joint structural dependencies that are critical for distinguishing benign activity from coordinated lateral movement. Therefore, our objective is to learn representations that integrate both pairwise interaction signals and high-order contextual dependencies.

This problem setting is particularly challenging due to the dynamic nature of enterprise networks, the partial observability of interactions, and the need to generalize beyond entities observed during training.

B. High-order Temporal Interaction Modeling

A central limitation of existing temporal graph formulations lies in their reliance on pairwise decomposition. In enterprise

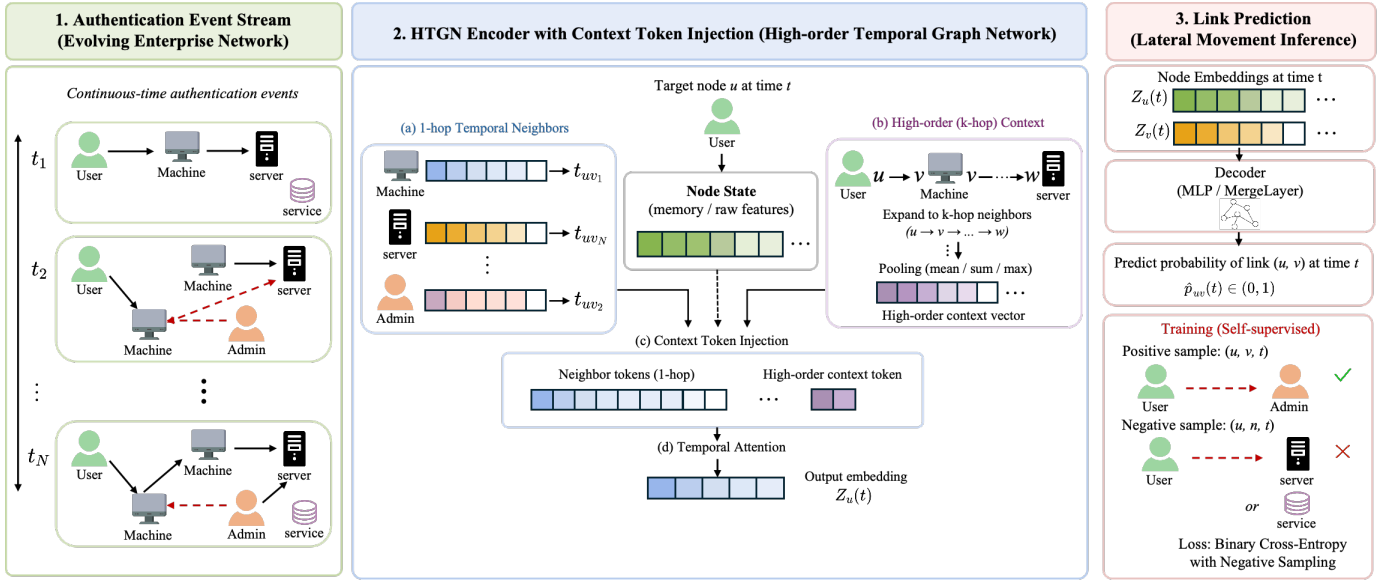


Fig. 2. Overview of the proposed framework for lateral movement inference in evolving enterprise networks. Authentication events are modeled as dynamic interactions among multiple entities. The encoder captures both temporal neighbors and high-order contextual information to learn enriched node representations, which are then used for temporal link prediction.

networks, authentication behaviors are rarely isolated interactions between two entities. Instead, a single event often involves multiple entities simultaneously, such as a user, a source host, a destination host, and associated services. Modeling such interactions as independent pairwise edges breaks their inherent joint structure and discards higher-order dependencies, leading to incomplete or ambiguous representations.

To address this issue, we adopt a high-order temporal interaction modeling framework. Let $\mathcal{V}$ denote the set of entities, and let $\mathcal{E} = \{e_1, e_2, \dots, e_T\}$ denote the sequence of timestamped events, where each event e_t corresponds to a set of participating entities at time t , i.e., $e_t \subseteq \mathcal{V}$. To capture the joint information within each event, we construct an event-level representation by aggregating the features of participating entities:

$$\mathbf{h}_{e_t} = \text{AGG}(\{\mathbf{h}_u^t \mid u \in e_t\}), \quad (1)$$

where $\text{AGG}(\cdot)$ denotes a permutation-invariant aggregation function and $\mathbf{h}_u^t$ is the representation of entity u at time t . In practice, this event-level representation is used as contextual information when computing node embeddings, allowing each interaction to be informed not only by pairwise relations but also by the broader set of entities involved in the same event.

This formulation preserves high-order structural dependencies without explicitly decomposing events into independent edges, while remaining compatible with the temporal link prediction objective. Specifically, predictions are still made over pairs of entities, but are conditioned on representations enriched with high-order contextual signals. As a result, the model can better capture coordinated and multi-stage behaviors that are characteristic of lateral movement in enterprise networks.

C. Temporal Attention with Smooth Normalization

To capture temporal dependencies in evolving interaction sequences, we employ an attention mechanism over historical events. For a target interaction at time t , we consider its temporal neighborhood $\mathcal{N}(t)$, which contains relevant past interactions. The attention weights are defined as

$$\alpha_i = \frac{\phi(s(\mathbf{q}_t, \mathbf{k}_i))}{\sum_{j \in \mathcal{N}(t)} \phi(s(\mathbf{q}_t, \mathbf{k}_j))}, \quad (2)$$

where $\mathbf{q}_t$, $\mathbf{k}_i$, and $\mathbf{v}_i$ denote the query, key, and value representations, and $s(\cdot, \cdot)$ is a similarity function. In standard formulations, $\phi(x) = \exp(x)$ corresponds to SoftMax normalization. However, prior studies [10] have shown that SoftMax is inherently constrained by a trade-off between sparsity and multi-modality, making it difficult to simultaneously preserve multiple relevant interactions while suppressing noise. This limitation is particularly pronounced in dynamic enterprise environments, where interactions are noisy and may involve multiple concurrent behavioral patterns. Inspired by this observation, we adopt a smooth alternative to exponential normalization by setting

$$\phi(x) = \text{Softplus}(x).$$

Compared to exponential mapping, the Softplus function provides a smoother transformation that avoids excessive amplification of large scores while preserving non-negativity. This results in a more balanced attention distribution, where multiple relevant interactions can retain meaningful contributions, and low-relevance signals are softly suppressed. Additionally, the smooth nature of Softplus leads to improved gradient stability during training, which is beneficial for modeling long-range

temporal dependencies. The aggregated representation is then computed as

$$\mathbf{c}_t = \sum_{i \in \mathcal{N}(t)} \alpha_i \mathbf{v}_i. \quad (3)$$

Empirically, this smooth normalization consistently improves performance over standard SoftMax, demonstrating its effectiveness in capturing complex temporal interaction patterns.

D. Training Objective

Building upon the learned temporal representations, lateral movement (LM) inference is formulated as a temporal link prediction problem, where the model estimates the likelihood that an interaction between entities at time t corresponds to a potential lateral movement step. Let $\mathbf{h}_u^t$ and $\mathbf{h}_v^t$ denote the representations of entities u and v at time t , which are obtained from the high-order temporal encoding. We define the interaction score as

$$\hat{y}_{uv}^t = \sigma(g(\mathbf{h}_u^t, \mathbf{h}_v^t)),$$

where $g(\cdot)$ is a learnable scoring function and $\sigma(\cdot)$ is the sigmoid function. The model is trained using a binary cross-entropy objective to distinguish observed interactions from unobserved ones:

$$\mathcal{L} = - \sum_{(u,v,t) \in \mathcal{D}^+} \log \hat{y}_{uv}^t - \sum_{(u,v,t) \in \mathcal{D}^-} \log (1 - \hat{y}_{uv}^t),$$

where $\mathcal{D}^+$ and $\mathcal{D}^-$ denote the sets of positive and negative samples, respectively. Negative samples are generated by randomly corrupting one endpoint of observed interactions, enabling the model to distinguish plausible interactions from spurious ones.

To support inductive inference, training is conducted on a subset of observed entities and interactions, while evaluation is performed on unseen nodes and future events. This setting reflects the dynamic and partially observed nature of enterprise networks and evaluates the model's ability to generalize to new entities and evolving interaction patterns.

IV. EXPERIMENTS

A. Experimental Setup

We evaluate the proposed framework on the Los Alamos National Laboratory (LANL) [20] authentication dataset, a widely used benchmark for lateral movement detection in enterprise networks. The dataset consists of time-stamped authentication events among heterogeneous entities, including users, computers, and services, and contains labeled red-team activities that serve as ground truth for malicious lateral movement behaviors. Following prior work, we formulate the task as temporal link prediction over authentication interactions, where the objective is to estimate whether a future or unseen interaction corresponds to a lateral movement step.

To ensure a fair comparison, we follow the same inductive evaluation protocol as Jbeil. Specifically, the entity set is partitioned into seen and unseen subsets, where the model is

trained only on interactions involving seen entities, and evaluated on interactions that include unseen nodes. We consider three increasingly challenging settings, where 30%, 40%, and 50% of entities are held out during training, respectively. This protocol reflects realistic enterprise scenarios in which new users, hosts, or services may continuously emerge over time.

We compare our method against Jbeil, a strong temporal graph-based inductive baseline evaluated on the same dataset. Performance is measured using standard metrics for temporal link prediction and detection, including AUC, average precision (AP), precision, and recall. All experiments are implemented in PyTorch and conducted on a single NVIDIA RTX 4090 GPU. Unless otherwise specified, we use the Adam optimizer with a learning rate of 1×10^{-4} , a batch size of 200, and an embedding dimension of 100. We adopt early stopping with a patience of 5, sample up to 10 temporal neighbors for each interaction, and optimize the model using binary cross-entropy loss.

B. Overall Performance

We evaluate the overall performance of our method under inductive settings and compare it against a strong temporal graph-based baseline, Jbeil, using the same dataset and evaluation protocol. The results are reported in Table I. Across all inductive scenarios, our approach consistently outperforms the baseline in terms of AUC, average precision (AP), precision, and recall. Notably, the performance gains become more pronounced as the proportion of unseen nodes increases, indicating that the proposed framework generalizes more effectively in highly dynamic and partially observed environments.

Compared to conventional pairwise graph modeling, the proposed high-order interaction modeling captures richer structural dependencies among multiple entities within each event, leading to more informative representations. In addition, the smooth attention normalization improves the model's ability to suppress irrelevant interactions while preserving multiple meaningful dependencies. These two components jointly enhance the model's robustness and discriminative power in temporal link prediction. Overall, the results demonstrate that our approach provides a more expressive and effective framework for modeling complex temporal interactions in evolving enterprise networks.

C. Ablation Study

To better understand the contribution of each component, we conduct an ablation study focusing on two key aspects of our framework: (1) high-order structural modeling and (2) attention normalization. Unless otherwise specified, all ablation results are reported under the 40% unseen setting.

Impact of High-order Structural Modeling. We evaluate the effect of incorporating high-order structural information into temporal graph modeling. Specifically, we compare the baseline setting ($k = 1$), which only considers 1-hop temporal neighbors, with higher-order variants ($k \geq 2$) that additionally capture multi-hop dependencies. As shown in Table II, introducing high-order context consistently improves performance

TABLE I
OVERALL PERFORMANCE UNDER INDUCTIVE SETTINGS ON THE LANL DATASET.

Method	30% Unseen				40% Unseen				50% Unseen			
	AUC	AP	Prec	Rec	AUC	AP	Prec	Rec	AUC	AP	Prec	Rec
Jbeil [9]	0.9973	0.9961	0.9925	0.9925	0.9948	0.9932	0.9881	0.9876	0.9915	0.9894	0.9823	0.9817
Ours	0.9981	0.9973	0.9941	0.9938	0.9962	0.9950	0.9906	0.9901	0.9938	0.9921	0.9857	0.9849

TABLE II
ABLATION STUDY ON HIGH-ORDER DEPTH AND POOLING STRATEGIES.

k	Pooling	AUC	AP	Prec	Rec
1	Mean	0.9948	0.9932	0.9881	0.9876
2	Mean	0.9959	0.9943	0.9904	0.9900
2	Sum	0.9955	0.9942	0.9896	0.9890
2	Max	0.9958	0.9945	0.9900	0.9894
3	Mean	0.9953	0.9936	0.9901	0.9896

across all evaluation metrics. In particular, the configuration with $k = 2$ achieves the best overall performance, indicating that incorporating 2-hop structural information provides the most effective balance between informative context and noise. We further observe that different pooling strategies exhibit varying behaviors. Mean pooling achieves the best overall performance, while max pooling slightly improves AP, suggesting that emphasizing the most salient interactions can be beneficial for certain metrics. When increasing k to 3, the performance slightly degrades, indicating that excessively large receptive fields may introduce noise and reduce representation quality. These results demonstrate that modeling local multi-hop dependencies is crucial, while overly deep structural expansion is less effective. In particular, lateral movement behaviors in enterprise networks often involve multi-step propagation across entities, which cannot be fully captured by 1-hop temporal interactions alone.

TABLE III
ABLATION STUDY ON ATTENTION NORMALIZATION.

Normalization	AUC	AP	Prec	Rec
SoftMax [9]	0.9948	0.9932	0.9881	0.9876
MultiMax [10]	0.9958	0.9947	0.9899	0.9896
Ours	0.9960	0.9949	0.9903	0.9899

Impact of Attention Normalization. We further examine the effect of attention normalization by replacing the proposed Softplus-based MultiMax with the standard SoftMax formulation. As shown in Table III, the model with SoftMax exhibits consistently lower performance across all evaluation metrics. This suggests that relaxing the strict competition imposed by SoftMax allows the model to better capture multiple relevant interactions in temporal contexts. In particular, the proposed smooth MultiMax formulation improves training stability and provides more flexible weighting over historical

interactions, leading to more robust representations in dynamic environments. Overall, the ablation results indicate that both high-order structural modeling and attention normalization contribute to the performance improvements, and their combination yields the best results.

V. CONCLUSION

In this paper, we studied the problem of lateral movement inference in evolving enterprise networks and identified two key limitations of existing temporal graph-based approaches: the insufficient modeling of high-order interaction structures and the limitations of SoftMax-based attention normalization. To address these issues, we proposed a high-order temporal modeling framework that captures multi-entity interactions at the event level, together with a smooth normalization mechanism for temporal attention. These designs enable more expressive representations of complex dependencies and provide more balanced weighting over historical interactions. Experimental results on real-world enterprise network data demonstrate that the proposed approach consistently improves over strong temporal graph-based baselines under inductive settings. Further analysis shows that both high-order structural modeling and attention normalization contribute to the overall performance gains.

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